

Antonio Álvarez-López

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Curriculum Vitae

Profile

Machine learning researcher with a PhD in Mathematics, working on the mathematical foundations and computational applications of neural ODEs, transformers, generative models, and time-series analysis. Incoming Postdoctoral Researcher at **Nanyang Technological University (NTU)**.

Education

Dec 2022–Jun 2026 **PhD in Mathematics**, *Universidad Autónoma de Madrid (UAM)*, Madrid, Spain.
Thesis: *Deep Learning with Controlled Flows: Expressivity, Generalization and Generation*.
Advisors: Enrique Zuazua Iriondo, Rafael Orive Illera. **Cum Laude, International Mention**

2021–2022 **M.Sc. in Advanced Mathematics**, *Universidad Complutense de Madrid*, Madrid, Spain.
Thesis: *Parker's Problem in Magnetohydrodynamics and transition maps for force-free fields*.
Advisor: Alberto Enciso.

2016–2021 **Bachelor's double degree in Mathematics and Physics**, Universidade de Santiago de Compostela (USC), Spain, Academic year 2019–2020 in Universidad de Granada.
Highest Honors in 19 out of 59 courses.

Experience

Sep 2026– **Incoming Postdoctoral Researcher**, *School of Physical and Mathematical Sciences, NTU*, Singapore.
Supervisors: J.-P. Ortega and Florian Rossmannek.

Apr 2024–Apr 2026 **Research Associate**, *Chair for Dynamics, Control, Machine Learning and Numerics (FAU-DCN), Friedrich-Alexander-Universität Erlangen–Nürnberg*, Erlangen, Germany.

Jun–Jul 2021 **Machine Learning Research Intern**, *CiTIUS*, Santiago de Compostela, Spain.
Advisor: Paulo F. Lamas. Project: Pattern recognition in cardiovascular diseases using machine learning.

Jul–Sep 2020 **Severo Ochoa–ICMAT Research Intern**, *ICMAT*, Madrid, Spain.
Advisor: Alberto Enciso. Project: Mathematical problems in fluid mechanics.

Research Visits

Aug–Nov 2025 **National University of Singapore (NUS)**, Singapore.
Host: Qianxiao Li.

May–Jul 2025 **Sorbonne Université, Laboratoire Jacques-Louis Lions**, Paris, France.
Host: Borjan Geshkovski.

2022–Present **FAU-DCN, Friedrich-Alexander-Universität Erlangen–Nürnberg**, Erlangen, Germany.
Recurrent visits, approximately two months per year. Host: Enrique Zuazua.

Jan 2024 **Sichuan University, School of Mathematics**, Chengdu, China.
Host: Qi Lü.

Aug–Sep 2023 **Massachusetts Institute of Technology, Department of Mathematics**, Cambridge, USA.
Host: Philippe Rigollet and Borjan Geshkovski.

Feb–Mar 2023 **Imperial College London, Department of Mathematics**, London, United Kingdom.
Host: Domènec Ruiz-Balet.

Papers

Published and Accepted

2026 **Gaussian mixture models in Hilbert spaces via kernel methods**, *Daniel López-Montero**, ***Antonio Álvarez-López****, *Marcos Matabuena*, Advances in Neural Information Processing Systems (NeurIPS) 2026. (*Equal contribution). arXiv:2605.05996.

2026 **Perceptrons and localization of attention's mean-field landscape**, ***Antonio Álvarez-López***, *Borjan Geshkovski, Domènec Ruiz-Balet*, International Conference on Machine Learning (ICML) 2026.
Spotlight paper (top 2.2% of all submissions). arXiv:2601.21366.

2025 **Constructive approximate transport maps with normalizing flows**, ***Antonio Álvarez-López***, *Borjan Geshkovski, Domènec Ruiz-Balet*, Applied Mathematics and Optimization, 92, 33.
DOI: 10.1007/s00245-025-10299-7.

2025 **Cluster-based classification with neural ODEs via control**, ***Antonio Álvarez-López***, *Rafael Orive-Illera, Enrique Zuazua*, Journal of Machine Learning, 4(2), 128–156.
DOI: 10.4208/jml.241114.

2024 **Interplay between depth and width for interpolation in neural ODEs**, ***Antonio Álvarez-López***, *Arselane Hadj Slimane, Enrique Zuazua*, Neural Networks, 180, 106640.
DOI: 10.1016/j.neunet.2024.106640.

Submitted

2026 **Constructive interpolation and generalization rates for neural ODEs: A control perspective**, ***Antonio Álvarez-López***, *Lorenzo Liverani, Enrique Zuazua*.
arXiv:2606.00469.

2026 **Continuous-time learning of probability distributions: A case study in a digital trial of young children with type 1 diabetes**, ***Antonio Álvarez-López***, *Marcos Matabuena*.
Best Student Paper Award. arXiv:2603.24427.

2025 **Convergence, design and training of continuous-time dropout as a random batch method**, ***Antonio Álvarez-López***, *Martín Hernández*.
arXiv:2510.13134.

Selected Talks and Presentations

Invited Seminars

Jun 18, 2026 *Perceptrons and localization of attention's mean-field landscape*. Math Machine Learning Seminar, **MPI MiS and UCLA**.

May 14, 2026 *Deep Learning with Controlled Flows: Expressivity, Generalization and Generation*. Machine Learning Seminar, **Universidad Autónoma de Madrid**.

Sep 29, 2025 *Entropy-driven control of the continuity equation for normalizing flows*. Machine Learning and Scientific Computing Seminar, **National University of Singapore**.

Sep 16, 2025 *Entropy-driven control of the continuity equation for normalizing flows*. Nonlinear PDE Seminar, Department of Mathematics, **Texas A&M University**.

Jul 9, 2025 *New deep learning models and perspectives for continuous-time glucose monitoring*. DeustoCCM Seminar, **University of Deusto, DeustoTech**.

2024–2025 Lecture series presentations in *Learning Rate — Machine Learning Series*, 4 sessions, **FAU-DCN**.

Apr 1, 2024 *Controllability of neural differential equations*. Seminario Fontán de Matemáticas, **USC**.

Feb 22, 2024 *Stable Architectures for Deep Neural Networks*. Mathematics for Deep Learning Reading Group, **Universities of Bath, Cambridge, and University College London**.

Jan 25, 2024 *Efficient classification and interpolation with neural ODEs*. Mathematics Seminar, School of Mathematics, **Sichuan University**.

Mar 1, 2023 *On the separability of two finite sets of random points using hyperplanes*. Mini-workshop between **Technische Universität München** and **FAU-DCN**.

Feb 28, 2023 *Breaking the curse of dimensionality with Barron Spaces*. Applied PDE Seminar, **Imperial College London**.

Nov 14, 2022 *Breaking the curse of dimensionality with Barron Spaces*. **FAU-DCN**.

Apr 4, 2021 *Formation of singularities in the incompressible Euler fluid equations*. Seminarios de Iniciación a la Investigación, Instituto de Matemáticas, **USC**.

Conference Presentations

Jul 9, 2026 **Poster**, *International Conference in Machine Learning*, Seoul, South Korea.
Perceptrons and localization of attention's mean-field landscape. **Spotlight paper**.

Jun 23, 2026 **Contributed Talk**, *Machine Learning and PDEs Workshop*, FAU MoD, Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany.
Perceptrons and localization of attention's mean-field landscape.

Jun 11, 2026 **Contributed Talk**, *Control, PDEs and Machine Learning: Conference in Honour of Enrique Zuazua's 65th Birthday*, University of Deusto, Bilbao, Spain.
Perceptrons and localization of attention's mean-field landscape.

Dec 2025 **Contributed Talk**, *International Conference on Statistics and Data Science*, Universidad de Sevilla, Spain.
Continuous-time learning of probability distributions via neural ODEs with applications in continuous glucose monitoring data. **Best Student Paper Award**.

Oct 2024 **Poster**, *Conference on Control, Inversion and Numerics for PDEs*, Fudan University, Shanghai, China.
Controllability of Neural ODEs for Classification.

Sep 2024 **Poster**, *Mathematical Aspects of Learning Theory — 20 Years Later*, Centre de Recerca Matemàtica, Barcelona, Spain.
Simultaneous Control of Neural ODEs for Data Classification.

Aug 2024 **Invited Talks**, *X Partial Differential Equations, Optimal Design and Numerics*, Centro de Ciencias de Benasque Pedro Pascual, Huesca, Spain.
Two invited talks: *Cluster-based classification with neural ODEs via control* and *Interplay between depth and width for interpolation in neural ODEs*.

Jul 2024 **Poster**, *Fourth Symposium on Machine Learning and Dynamical Systems*, Fields Institute, Canada.
Simultaneous Control of Neural ODEs for Data Classification.

Jun 2024 **Invited Talk**, *21st French-German-Spanish Conference on Optimization*, Universidad de Oviedo, Spain.
Controllability of neural ODEs for data classification.

Jun 2024 **Poster**, *Trends in Mathematical Sciences*, FAU, Erlangen, Germany.
Optimized Classification with Neural ODEs.

Mar 2024 **Poster**, *Dynamics, Data and Deep Learning Workshop*, Engineers' House, Bristol, United Kingdom.
Optimized Classification with Neural ODEs. **Best Poster Award**.

Dec 2023 **Poster**, *Mathematical Opportunities in Digital Twins*, George Mason University, Fairfax, USA.
Optimized Classification with Neural ODEs.

Nov 2023 **Invited Talk**, *V BYMAT Conference*, ICMAT, Madrid, Spain.
Neural ODEs and data classification.

Academic Service

Jan–May 2026 **Teaching**, *Statistics*, Degree in Environmental Sciences, UAM.

Jan–May 2025 **Teaching**, *Calculus II*, Degree in Data Science and Engineering, UAM.

Jan–May 2025 **Teaching**, *Probability and Statistical Inference*, Degree in Computer Science, UAM.

Feb–May 2024 **Teaching**, *Numerical Calculus*, Double degree in Mathematics and Computer Science, UAM.

Feb–May 2023 **Teaching**, *Data Analysis*, Degree in Biology, UAM.

2024–2025 **Co-organizer**, *Machine Learning Seminars and Junior Colloquia at UAM, and Lecture Group at FAU-DCN*.

Awards and Fellowships

2026 **ICML Spotlight Paper**, *Top 2.2% of submissions*.

2025–2026 **FPU Mobility Supplement (research visit at NUS)**, *Ministerio de Universidades*, Spain.
Ranked #1 nationwide.

2025 **Best Student Paper Award**, *2025 IMS International Conference on Statistics and Data Science (ICSDS)*.

2024 **Best Poster Award**, *Dynamics, Data and Deep Learning Workshop, Bristol*.

2022–2026 **FPU Predoctoral Fellowship**, *Ministerio de Universidades*, Spain.
Ranked #6 nationwide.

2021 **Research Fellowship "JAE Intro SOMdM 2021"**, *ICMAT, Madrid*.
Awarded to 7 students nationwide.

2018 **Academic Excellence Award**, *Best qualifications of the Galician University System*.

2016 **USC's A Ponte Award**, *Top students in the University Entrance Exam*.

2014 **National Award for Academic Achievement**, *Spanish Lower Secondary Education*.
Awarded to 25 students nationwide.

Technical Skills

Languages **Spanish (Native), Galician (Native), English (C2), French (B2), German (Basic).**

Programming **Python, Matlab, R, Fortran**